Option 3:
$\frac{96}{\sqrt{5}}$

Option 4:
12

Correct Answer:
$\frac{96}{\sqrt{5}}$

Solution:
Ellipse $\frac{x^2}{36}+\frac{y^2}{16}=1$

$$e_E=\sqrt{\frac{36-16}{36}}=\frac{\sqrt{20}}{6}$$

$$\text{Foci }=\left(6\cdot\frac{\sqrt{20}}{6},0\right),\left(-6\cdot\frac{\sqrt{20}}{6},0\right)=(\sqrt{20},0),(-\sqrt{20},0)$$

$$\text{Hyperbola }:\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$$

$$ae_H=\sqrt{20}$$

$$5a=\sqrt{20}=2\sqrt{5}$$

$$a=\frac{2}{\sqrt{5}}$$

$$b^2=a^2\left(e^2-1\right)$$

$$b^2=\frac{4}{5}(25-1)=\frac{96}{5}$$

$$\therefore \text{Length of Latus Rectum}=\frac{2b^2}{a}=2\cdot\frac{96}{5}\cdot\frac{\sqrt{5}}{2}=\frac{96}{\sqrt{5}}$$

Hence, the correct answer is option (3).

Q. 10 Let $f:R\to(0,\infty)$ be a twice differentiable function such that $f(3)=18$, $f^l(3)=0$ and $f(3)=4$, then $\text{Lim}\left[\log_e\left(\frac{f(2+x)}{f(3)}\right)^{\frac{18}{(x-1)^2}}\right]$ is equal to:

Option 1:
1

Option 2:
18

Option 3:
2

Option 4:
9

Correct Answer:
2

Solution:

Given:

$\lim_{x\to 1} \log\left(\frac{f(2+x)}{f(3)}\right)^{\frac{18}{(x-1)^2}}$

Since $f(3) = 18$,

$= \lim_{x\to 1} \frac{18}{(x-1)^2} \ln\left(\frac{f(2+x)}{18}\right)$

As $x \to 1$,

$\ln\left(\frac{f(2+x)}{18}\right) \to 0$ and $(x-1)^2 \to 0$

So form is $\frac{0}{0}$.

Applying the L'Hospital rule, we get,

$$= \lim_{x\to 1} \frac{18 \cdot \frac{f'(2+x)}{f(2+x)}}{2(x-1)}$$

$$= \lim_{x\to 1} \frac{9f'(2+x)}{(x-1)f(2+x)}$$

Since $f'(3) = 0$, still $\frac{0}{0}$.

Applying the L'Hospital rule again, we get,

Numerator derivative:

$9f''(2+x)$

Denominator derivative:

$f(2+x) + (x-1)f'(2+x)$

So,

$= \lim_{x\to 1} \frac{9f''(2+x)}{f(2+x)+(x-1)f'(2+x)}$

Substitute $x = 1$:

$= \frac{9f''(3)}{f(3)} = \frac{9\cdot 4}{18} = \frac{36}{18} = 2$

Hence, the correct answer is option (3).

Q. 11 The number of strictly increasing functions f from set $\{1, 2, 3, 4, 5, 6\}$ to the set $\{1, 2, 3, \ldots\ldots\ldots, 9\}$ such that $f(i) \neq i$ for $1 \leq i \leq 6$ is equal to:

Option 1:
21

Option 2:
27

Option 3:
22

Option 4:
28

Correct Answer:
28

Solution:
No. of functions $= 8_{C_6} = 8_{C_2} = 28$

Hence, the correct answer is option (4).

The weight of the liquid column of height h is :

$W = \left(\pi r^2 h\right)\rho g$

At equilibrium :

$\mathrm{T}\cos\theta(2\pi \mathrm{r}) = \pi \mathrm{r}^2\,\mathrm{h}\rho\,\mathrm{g}$
$\Rightarrow\ \mathrm{h} = \dfrac{2\,\mathrm{T}\cos\theta}{\mathrm{r}\rho\,\mathrm{g}}$

If we assume the two liquids refer to the same substance, then:
$\mathrm{T}_1 = \mathrm{T}_2 \ldots$ (i) (Surface tension is a property of the material and temperature).
ρ and g are constant.
θ is constant.
This results the relationship: $\mathrm{h} \propto \frac{1}{\mathrm{r}}$.
We are given that $r_1 > r_2 \Rightarrow \frac{r_1}{r_2} > 1$.
So, the ratio of the heights of capillary rise,

$\frac{\mathrm{h}_1}{\mathrm{h}_2} = \frac{\mathrm{r}_2}{\mathrm{r}_2} < 1 \Rightarrow\ \mathrm{h}_1 < \mathrm{h}_2 \ldots (ii)$
From both the relations, $\mathrm{h}_1 < \mathrm{h}_2$ and $\mathrm{T}_1 = \mathrm{T}_2$.

Q. 20 The energy of an electron in an orbit of Bohr's atom is $-0.04E_0\mathrm{eV}$ where E_0 is the ground state energy. If L is the angular momentum of electron in this orbit and h is planks constant then $\frac{2\pi L}{h}$ is ____ .

Option 1:
2

Option 2:
6

Option 3:
5

Option 4:
4

Correct Answer:
5

Solution:
Angular momentum $L = \frac{nh}{2\pi}$

$n = \frac{2\pi L}{h}$

Energy $E = \frac{-13.6}{n^2} \cdot z^2 = -0.04 \times \left(\frac{+13.6}{1} \cdot z^2\right)$

$n^2 = 25$
$n = 5$

Q. 21 A diatomic gas $(\gamma = 1.4)$ does $100J$ of work when it is expanded isobarically. Then the heat given to the gas is ____ J.

Correct Answer:
350

Solution:
$W = 100J = nR\Delta T$ for isobaric process .

$Q = nC_p\Delta T = \left(\frac{f}{2} + 1\right)nR\Delta T$

$1-2x=0 \Rightarrow x=\frac{1}{2}$

Parabola cuts x-axis at

$4-x^2=0 \Rightarrow x=2$

So required area:

$A=\int_0^2(4-x^2)dx-\int_0^{1/2}(1-2x)dx$

Now compute:

$\int_0^2(4-x^2)dx=\left[4x-\frac{x^3}{3}\right]_0^2=8-\frac{8}{3}=\frac{16}{3}$

$\int_0^{1/2}(1-2x)dx=\left[x-x^2\right]_0^{1/2}=\frac{1}{2}-\frac{1}{4}=\frac{1}{4}$

So, $A=\frac{16}{3}-\frac{1}{4}=\frac{55}{12}$

Thus, $\alpha=55, \beta=12$

$\alpha+\beta=67$

Hence, the correct answer is option (2).

Q. 3 A random variable X takes values $0,1,2,3$ with probabilities $\frac{2a+1}{30}, \frac{8a-1}{30}, \frac{4a+1}{30}$, b respectively, where $a, b \in R$. Let μ and σ respectively be the mean and standard deviation of X such that $\sigma^2+\mu^2=2$. Then $\frac{a}{b}$ is equal to:

Option 1:
3

Option 2:
30

Option 3:
12

Option 4:
60

Correct Answer:
60

Solution:

x	0	1	2	3
$p(n)$	$\frac{2a+1}{30}$	$\frac{8a-1}{30}$	$\frac{4a+1}{30}$	b

$\mu=\Sigma \times p(n)=0+\frac{8a-1}{30}+2\times\frac{4a+1}{30}+3b=\frac{16a+90b+1}{30}$

$\sigma^2=\Sigma x^2p(n)-\mu^2 \Rightarrow \sigma^2+\mu^2=\Sigma x^2p(n)=2$

$\frac{8a-1}{30}+\frac{4(4a+1)}{30}+9b=2$

$8a+90b-19=0 \ldots\ldots\ldots (i)$

$\Sigma p(n)=1 \Rightarrow 14a+1+30b=30$

$14a+30b-29=0 \ldots\ldots (ii)$

From (i) and (ii)

$a=2 \quad b=\frac{1}{30} \quad \frac{a}{b}=60$

Hence, the correct answer is option (4).

Q. 4 The positive integer n , for which the solutions of the equation $x(x+2)+(x+2)(x+4)+\ldots+(x+2n-2)(x+2n)=\frac{8n}{3}$ are two consecutive even integers, is:

Option 1:
3

Option 2:
9

Option 3:
6

Option 4:
12

Correct Answer:
3

Solution:

$$nx^2+(2x+6x+10x+\ldots\ldots+(4n-2))+(2.4+4.6+\ldots+(2n-2)2n)=\frac{8n}{3}$$

$$nx^2+2x(1+3+5+\ldots.+2n-1)+\sum_{r=1}^{n}(2r-2)2r=\frac{8n}{3}$$

$$nx^2+2n^2x+4\left(\frac{n(n+1)(2n-2)}{6}\right)=\frac{8n}{3}$$

$$3x^2+6nx+4\left(n^2-1\right)=8$$

For $\mathrm{n}=3$

$$3x^2+18x+24=0$$

$x=-2,-4$ (two consecutive even integers)

Hence, the correct answer is option (1).

Q. 5 Let $f(x)=x^3+x^2f'(1)+2xf''(2)+f'''(3), x\in R$. Then the value of $f'(5)$ is :

Option 1:
$\frac{2}{5}$

Option 2:
$\frac{117}{5}$

Option 3:
$\frac{657}{5}$

Option 4:
$\frac{62}{5}$

Correct Answer:
$\frac{117}{5}$

Solution:
Let $f'(1)=a$

Structure II (1-bromobicycloheptane): This is a more rigid and strained bridgehead system than structure I. The small rings prevent the bridgehead carbon from becoming even partially planar, making the carbocation formation extremely difficult and the reactant virtually unreactive in SN1 mechanisms.

Hence, the correct answer is the option (3).

Q. 10 Given below are two statements :

Statement I : Phenol on treatment with $\mathrm{CHCl_3}$ / aq. KOH under refluxing condition, followed by acidification produces p - hydroxy benzaldehyde as the major product and oohydroxy benzaldehyde as the minor product.

Statement II : The mixture of p-hydroxybenzaldehyde and o- hydroxybenzaldehyde can be easily separated through steam distillation.

In the light of the above statements, choose the correct answer from the options given below

Option 1:
Both Statement I and Statement II are false

Option 2:
Statement I is true but Statement II is false

Option 3:
Statement I is false but Statement II is true

Option 4:
Both Statement I and Statement II are true

Correct Answer:
Both Statement I and Statement II are true

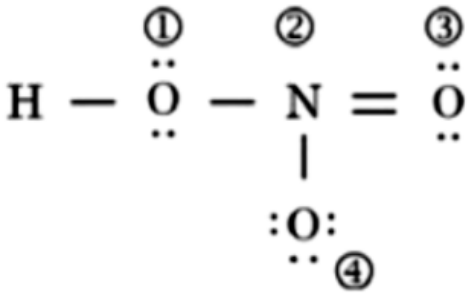
Solution:
Sol: Due to steric factor of K^+para product is major

Statement I: True — Phenol + $\mathrm{CHCl_3}$/aq. KOH → Reimer-Tiemann reaction, gives p-hydroxybenzaldehyde (major) and o-hydroxybenzaldehyde (minor).

Statement II: True — The two isomers can be separated by steam distillation.

Hence, the correct answer is the option (4).

Q. 11 The formal charges on the atoms marked as (1) to (4) in the Lewis representation of HNO_3 molecule respectively are



Option 1:
$0, 0, -1, +1$

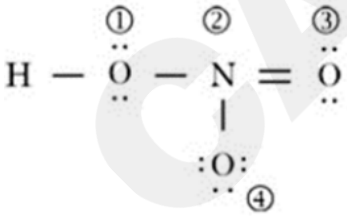
Option 2:
$0, -1, 0, +1$

Option 3:
$0, +1, 0, -1$

Option 4:
$+1, 0, 0, -1$

Correct Answer:
$0, +1, 0, -1$

Solution:



$1 = 6 - 4 - \frac{1}{2}[4] = 0$

$2. 5 - 0 - \frac{1}{2}[8] = +1$
$3. 6 - 4 - \frac{1}{2}[4] = 0$
$4. 6 - 6 - \frac{1}{2}[2] = -1$

Hence, the correct answer is the option (3).

$q_1 = +2qr_1 = -3a\hat{j}$

$q_1 = +3qr_2 = 2a\hat{i}$

$q_3 = +4qr_3 = -2a\hat{i}$

$P = \varepsilon qr = q_1r_1 + q_2r_2 + q_3r_3$

$P = (2q)(-3a\hat{j}) + 3q(2a)\hat{i} + (-4q)(-2a)\hat{i}$

$P = 2qa[7\hat{i} - 3\hat{j}]$

Q. 13 Two resistor's of 100Ω each are connected in series with a 9 V battery a voltmeter of 400Ω resistance is connected two measure the voltage drop a cross one of the resistance the voltmeter reading is _____ V

Option 1:
4.5

Option 2:
4

Option 3:
3

Option 4:
2

Correct Answer:
4

Solution:
Sol:

Req $= 100 + 800 = 180$

$I = \frac{9}{180}$

$\text{Voltmeter reading } = iR = \frac{9}{180} \times 80 = 4v$

Q. 14 Two masses 400 g and 350 g are suspended from the ends of a light string passing over a heavy pulley of radius 2 cm when released from rest the heavier mass is observed to fall 81 cm is 9 s . The rotational inertia of the pulley is _____ $kg.m^2.\left(g = 9.8\ m/s^2\right)$

Option 1:
4.75×10^{-3}

Option 2:
1.86×10^{-2}

Option 3:
9.5×10^{-3}

Option 4:
8.3×10^{-3}

Correct Answer:
9.5×10^{-3}

Solution:
Sol:

$a = \frac{(m_2 - m_1)g}{\left[m_1 + m_2 + \frac{I}{R^2}\right]} \Rightarrow S = 21t + \frac{1}{2}at^2$

$$\frac{81}{100} = 0 + \frac{1}{2}\left[\frac{m_2 - m_1}{m_1 + m_2 + \frac{I}{R^2}}\right] g \, (()^{\chi}$$

$$\rightarrow \left[m_1 + m_2 + \frac{I}{R^2}\right] = \frac{1}{2}\frac{(m_2 - m_1)(g)}{2}100$$

$$\left[\frac{750}{1000} + \frac{I}{R^2}\right] = \frac{1}{2}\left[\frac{50}{1000}\right](9.8)100$$

$$I = 9.5 \times 10^{-4} \cdot \text{ kg m}^2$$

Q. 15 A boy throw's a ball in to air at 45° from the horizontal to land on a roof of a building of height H . If the ball attains maximum height in 2 s and land on the building in 3 s after launch then the value of H is _____ $m \left(g = 10 \text{ m/s}^2\right)$

Option 1:
15

Option 2:
20

Option 3:
10

Option 4:
25

Q. 7 An organic compound undergoes first order decomposition. The time taken for decomposition to $\left(\frac{1}{8}\right)^{\text{th}}$ and $\left(\frac{1}{10}\right)^{\text{th}}$ of its initial concentration are $t_{1/8}$ and $t_{1/10}$ respectively. What is the value of $\frac{t_{1/8}}{t_{1/10}} \times 10$?

$(\log 2 = 0.3)$

Option 1:
0.9

Option 2:
30

Option 3:
3

Option 4:
9

Correct Answer:
9

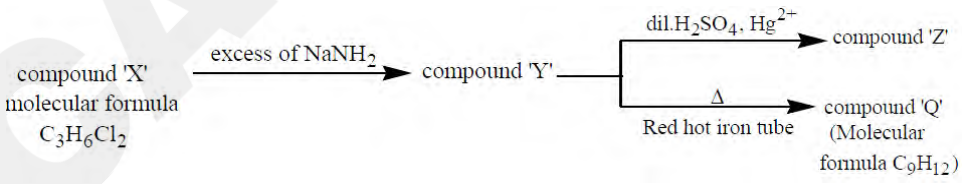
Solution:

$$\frac{t_{1/8}}{t_{1/10}} = \frac{\frac{2.303}{K}\log\frac{[A_O]}{1/8[A_O]}}{\frac{2.303}{K}\log\frac{[A_O]}{1/10[A_O]}} = 0.9$$

$$\frac{t_{1/8}}{t_{1/10}} \times 10 = 9$$

Hence, the correct answer is option (4)

Q. 8 Given below are two statements for the following reaction sequence.



Statement I : Compound 'Z' will give yellow precipitate with NaOI.
Statement II: Compound 'Q' has two different types of 'H' atoms (aromatic : aliphatic) in the ratio 1 : 3
In the light of the above statements, choose the correct answer from the options

Option 1:
Statement I is true but Statement II is false